

WRITTEN REPORT FOR A HIGHER DEGREE BY THESIS	
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Tipping points, the topic of Joshua's thesis, are of great relevance in mathematical, natural, social, and applied sciences, and delivering a PhD thesis on this subject is extremely timely. Joshua is particularly interested in studying possible early warning indicators, in order to provide tools for anticipating the occurrence of tipping points and reducing their impacts in a variety of situation of social and environmental interest. We comment below on aspects that the committee has found especially critical in the thesis and in the discussion during the viva, which has confirmed strengths and weaknesses of the thesis. While the debate on tipping points for the Earth system is extremely prominent in geosciences, very little is reported in the thesis. This needs to be expanded, as Joshua's results need to be put in a broader context. Additionally, the three kinds of tipping behaviours need to be described more thoroughly. The three indicators discussed by Joshua rely heavily on the concept of scaling behaviour. A fundamental issue associated to using these methods is the need for long datasets (and large ensembles thereof). Additionally the presence of periodic components can make these indicators (especially the DFA) useless. Instead, most of the signals investigated by Joshua are pretty short and the ensembles are rather small. The sensitivity of the results to the length of the signals and the size of the ensemble should be investigated. Additionally, the mathematical reasons behind the expected scaling behaviour should be more clearly expressed and discussed. Finally, Joshua, oddly, does not provide a definition of the periodogram, which is one of the indicators The thesis is extremely unclear regarding the way noise is implemented in the dynamical systems of interest; the discussion with the candidate further clarified that his understanding of the numerics for stochastic differential equations is a bit weak, as a result of some confusion in some basic ideas behind Brownian motion. This aspect needs to be reinforced in the revised version of the thesis. Something that puzzled the committee is the fact that the geophysical application (presented in Chapter 4) of the previously described mathematical tools deals with a system that does not correspond to the description of tipping point. Indeed, tropical cyclones, whilst being very special and unique features of the atmosphere, do not conform to the general case of bistable system. Joshua needs to explain and motivate better the choice of this specific system of interest. Some of the results of the Chapter (e.g. the sensitivity of the results to using 99 vs 100 points as window) are unconvincing. A last missing aspect of Joshua's thesis is the absence of a final chapter giving the summary of the results, putting them into a broader perspective, and proposing future research lines. The committee has appreciated the attitude of the candidate during the interview. He has been extremely open, intellectually honest, and able to engage while being obviously stressed and under pressure. The two preliminary reports are attached to this document.	

Report 1

Title: Tipping points and early warning signals

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Supervisor: Tobias Kuna and Valerie Livina

Summary:

In his thesis, Joshua developed methods to study early warning signals (EWS) in (geophysical) time series. Joshua's assumption is that these signals are associated with tipping points. The thesis contains three main chapters, 2-4. In chapter 2 Joshua study/develop one dimensional tipping point techniques. In chapter 3, Joshua extends the one dimensional techniques into multiple variables and two dimensional technique. In chapter 4, Joshua applies the developed techniques on a specific geophysical time series, namely tropical cyclones (sea level pressure and wind speed).

The early warning techniques that Joshua uses include, the Fourier power spectrum (PS) exponent, the detrended fluctuations analysis (DFA) exponent, and the lag-1 auto-correlation (ACF1). The PS indicator is first used in Joshua's thesis. The different techniques exhibit reasonable performance when applied on synthetic examples/models (chapters 2 and 3). Yet, the performance is relatively poor when applied to the sea-level pressure and wind speed data of tropical cyclones.

Throughout the thesis, Joshua provides the mathematical details and rationale behind the developed techniques.

General conclusion:

The thesis is clearly written and well presented. In my eyes, chapter 3 contains the most interesting results and analysis. It is clear that Joshua performed deep and serious investigation of early warning signals. It is also clear that he has the ability to study in depth, numerically and analytically, the problem of early warning signal in relation with tipping points. It is unfortunate that the methods he developed are currently not applicable to serve as early indicators for significant climate shift, at least in the context of tropical cyclones. Yet, I think that the analysis and results presented in the thesis are sufficient to award Joshua a PhD. I have some general comments regarding the results and the methodology presented in the thesis and other, more minor, comments. These are listed below.

General comments:

1) Two of the early warning indicators used in the thesis are based on "scaling exponents", the PS and DFA scaling exponents. In addition, the scaling exponent of the auto-correlation function is also considered. Shortly, scaling is linked to power-law behavior in which there is no specific characteristic scale such that $f(x)=f(cx)/c^\alpha$ where α is the scaling exponent. Unfortunately, this seems to be overlooked, as the scaling exponent was estimated even when it is clear that there is no scaling, like when there is crossover (AR(1) process) and when the time series is "oscillatory" (like the Van-der-Pol oscillator). Indeed, it is possible to estimate the an exponent in a range where there seems to be a power-law behavior but, from what I understood, the estimation of the scaling exponent was done without checking if indeed this is the case. Even in the case of tropical cyclones where the author noted that there is a clear tidal 12 hours frequency and even proposed ways to filter out this frequency, eventually this frequency was not filtered out. In the last case, it is not clear that indeed there is scaling (probably not). The above may underlies the weak performance of the indicators. There are other technicalities related to the scaling which I list below.

2) The term "potential" is frequently used in the thesis. It seems to be used in inaccurate way. In physics "potential" is tightly linked to force through the relation $F = -\text{grad}(U)$ where the force is proportional to the second derivative of the location vector. In this context, a particle can be trapped in a potential well if its energy is smaller than the barrier's energy. However, in the examples considered in the dissertation, the potential is proportional to the first (time) derivative of the potential variable, e.g., $dx/dt = -dU/dx$. The "well" of U is not a real well as there is no relevant energy below which the "particle" is trapped. I would suggest the author to use the term "pseudo-potential" and to clarify that the potential he uses is not the same as the one used in physics in which a "well" is a well and a "barrier" is a barrier.

3) It is not clear that indeed tropical cycles belong to the class of tipping point mechanism. Please explain how this major example related to the tipping point discussed earlier.

Specific comments:

1) Page 6, 1st line after eq. (1.11): typo. Change to ϕ_1 .

2) Page 7, 4-5 after eq. (1.12): be more accurate here. The ACF is zero for $l \neq 0$ and for infinite series length.

3) Page 7, Fourier transform: is the transform combination of sine AND cosine functions?

4) Page 8, 1st line: Is the power spectrum actually delta function?

5) Page 8, 3 lines from above: it will be good to mention that the PS exponent of a random walk is -2 and that the PS and the ACF are related to each other through the Parseval's identity. You can also explain simply why the PS exponent of a RW is -2 by applying FT on $dx/dt = \eta$.

6) Eq. (1.13) and other locations. Note that this is essentially 1d ode as r does not depend on θ . Better to use real coupled system.

7) Eq. (1.14): note that this equation (and other similar equations throughout the thesis) are basically "parameterically-forced" system. Such systems have been investigated in the past and it is worth (in my eyes) mentioning this.

8) Page 9: you mention the term "bifurcation". Since it is a fundamental term used in the thesis I suggest to detail here the main different types of bifurcations, like sub-critical and super-critical.

9) Page 13: the last sentence on the page is not clear.

10) Page 18: It will be good to mention here the limitations of the different techniques (some of them are mentioned later in the text though). For example, ACF is sensitive to trends and the estimated scaling exponent has restricted range (γ is between 0 and 1). The PS is sensitive to trends. The scaling exponents of the DFA and ACF are extremely inaccurate when the time series exhibits oscillations.

11) Page 19: the reference to Kantenhardt et al. (2001): I believe that a more accurate reference will be Makse et al. (1994 or so). Also note that the original DFA reference is Peng et al. (1994 or so)--Kantenhardt et al. (2001) proposed the multifractal DFA which is not considered here.

12) Page 19 and Fig. 2.1: It is essential to use logarithmic binning in all techniques when estimating the scaling exponent, to avoid bias toward high frequency (small window scales). Also the range at which the scaling exponent is estimated is not fixed in all problems and should be checked carefully in every setup and problem. Scaling exponents are highly sensitive to this choice, specially when the series length is short.

13) Eq. (2.3): The AR(63) is unnecessary when generating red-noise as $a_2=a_3=\dots=0$ for $\lambda=2$. Only $a_0=1$ and $a_1=-1$ are needed. Red-noise series is basically RW series.

14) Page 13 (and elsewhere)--the ACF1 measure. Choosing always lag-1 independently of the temporal resolution (Δt) is problematic. For example, if Δt is very small, the ACF1 will be very close to 1 without real change even very close to the tipping point. If, on the other hand, Δt is very large the changes may be very large as the ACF1 may reach the fluctuation region where it fluctuates around zero. In other areas of science the lag was chosen according to some criteria, like the lag at which the ACF drops below 0.5 or so. I suggest the author to adopt such a criteria. This might be the reason for the relatively high ACF for the tropical cyclone time series.

15) Page 22: note (again) that the range for estimating the scaling exponent depends on the series and cannot be fixed.

16) Page 23, table 2.1: See comment 13 above. The Brown noise and red AR(63) are essentially the same...

17) Page 23: it will be good to mention the relations between the different scaling exponents and the scaling exponent of the ACF. I.e., $\beta=2\alpha-1=1-\gamma$ where γ is the scaling exponent of the ACF. See Maske et al. (1994 or so). Note that γ is limited between 0 and 1.

18) Eq. (2.12): note that there is no (real) scaling for AR(1) process as it exhibits crossover from flat spectrum at the low frequency to power-law drop (with exponent -2) for the high frequency. This can explain the deviations of Fig. 2.3a and this is related to the general comment regarding the scaling exponent mentioned above.

19) Page 26, 2nd paragraph: this make sense as there is no scaling.

20) Page 26, last paragraph: $\lambda=2$ is related red-noise and not $\lambda=1$. By red-noise I'm referring to $1/\omega^2$ PS. Is $\lambda=1$ the transition to non-stationary series?

21) Page 30. The relation between the different scaling exponents holds when there is real scaling--this is not the case here as obviously there are crossovers, and since it is not clear whether there is scaling or not.

22) Bottom of page 31: strictly speaking there might be "jumps" between the different segments. Better use eqs. (2.3) and (2.4) where λ vary with time continuously.

23) Page 36, last paragraph: again, see comment 21 above.

24) The constant in eq. (3.3) and eq. (3.4) will result in trend that will disturb the calculation of some of the indicators. Also note (again) that AR(1) process has a crossover and thus does not follow scaling laws.

25) One line after eq. (3.21): should N be replaced by P ?

26) Page 46: definition 3.2.3: this probably will result in biases as the length is not equal.

- 27) Eqs. (3.28) and (3.33): see a similar comment above. This is essentially 1d system.
- 28) Bottom of page 54 and Fig. 3.6: DFA scaling exponent does not have a meaning for an oscillatory series.
- 29) Eq. (3.85): You chose λ to be diagonal and to introduce the coupling through the noise via S . Why? Usually the variables themselves are coupled and the noise is specific to each variable separately.
- 30) Fig. 3.14: the value of the DFA exponent is very high, much higher than that of RW (1.5). Why? Is it due to inaccurate range at which the exponent was estimated?
- 31) Eq. (3.114): should c be normalized by the number of point or so (like $N(N-1)/2$)? The resulted values appears to be in the range of -1 to 1...
- 32) Page 89, 1st paragraph: what kind of tipping are you referring to?
- 33) Page 95, one sentence before 4.2. See comment above--not removing the periodicity can lead to problem in estimating the DFA exponent. Can you plot Fig. 4.2b in a log-log plot to see whether there is scaling?
- 34) Fig. 4.6. The range of the exponent is very large.
- 35) Page 106. The series from which the DFA and PS are calculated is very short. One cannot really estimate the scaling exponent from such a short series.
- 36) Page 110, one sentence after "Dimension reduction using EOFs". Delete the extra "the".
- 37) Eq. (4.3): Is a normalization factor missing?
- 38) Bottom of page 114: this questions the analysis.
- 39) Can you explain the need of EWS for tropical cyclones---there are good data/models that predict the development of the cyclones in time, days in advance.
- 40) Fig. 4.13: The difference between the different hurricanes is very large such that it is not clear whether there is a common EWS.
- 41) How did you get Eq. (4.7)? Is it related to $1/\rho(dp/dr)$?
- 42) Eq. (4.8): do you mean $2\pi(t+K)/12$?
- 43) Top of page 122: the changes in the PS seems to be tailored into the model such that the results are expected.
- 44) Page 122, first paragraph of the Discussion section. The increase is not so significant.
- 45) Top of page 125: the claim is valid for true scaling which is not the case here.
- 46) A concluding chapter is missing.

Report 2

Joshua's thesis deals with a very prominent scientific problem in the area of complex system science; namely, the properties of tipping points and the methods that can be used to anticipate their occurrence. Tipping points – or critical transitions, as they are more commonly referred to in physical and mathematical sciences – play an especially important role in geophysical sciences. They are intensely studied as extreme – and extremely relevant – cases of nonlinear response of the climate system to perturbations. After a tipping is realised, the system performs an irreversible changeover into a new state. Understanding the properties of tipping points and being able to detect/anticipate them from time series is clearly a key challenge of science. A fundamental issue in this research area is the desire to be able to study tipping points using one or low dimensional signals, whereas the systems one is typically interested in are extremely high (or infinitely) dimensional.

Joshua's thesis has the aim of summarizing some of the main results of proposing novel ideas in this research area. In particular, he aims at comparing the performance of a new early warning indicator (periodogram) against two indicators that have been widely discussed in the literature (lagged autocorrelation and exponent describing the properties of the detrended fluctuation analysis). Strangely Joshua does not provide a definition of the periodogram in his thesis. The thesis, in general, lacks precision in many points. In particular, there is no mention of how integration of stochastic differential equations are performed, little discussion of the choice of parameters in many toy models (rate of change of parameters; intensity of noise), nor clearly stated criteria for defining that the early warning indicator indeed flag the occurrence of a tipping point. The discussion of EOF analysis is, instead, quite accurate.

Something I find surprising is associated with the overall logical development of the thesis in its final part, and specifically, in the application to geophysical data. The previous part of the thesis describes the early warning indicators in relation to specific scenarios of tipping points (closely associated to bifurcations of low-dimensional systems). This is in line with part of the literature on the topic. The application of the theory takes place on pressure and wind data of tropical cyclones, which, as correctly stated by Joshua, have nothing to do with bifurcation, but he still classifies as “tipping behaviour”, with no further discussion. I believe that a logical step is missing there.

Finally, the thesis does not have a conclusions section, nor a sufficiently detailed discussion of tipping points in the Earth system. I believe that major revisions are needed before Joshua can be proclaimed Doctor in Philosophy.